

STATEMENT OF WORK (SOW)
HVAC System Replacement and Upgrade – Refrigerated Air Conversion
USDA Forest Service, Lincoln National Forest
Smokey Bear Ranger District, Ruidoso, New Mexico

1.0 PROJECT OBJECTIVE

The purpose of this project is to evaluate, design and replace existing HVAC systems in designated buildings with modern, energy efficient systems. The work includes system sizing, removal of outdated equipment, installation of new units, ductwork repair, electrical upgrades, and decommissioning of obsolete evaporative cooling systems.

The work will improve occupant comfort, system reliability, and climate control while complying with all current applicable codes, standards, and manufacturer requirements.

The Contractor shall provide all labor, materials, equipment, supervision, and incidentals necessary to deliver fully operational HVAC systems.

2.0 APPLICABLE CODES AND STANDARDS

All work shall comply with the most current versions of applicable codes and standards, including but not limited to:

- International Mechanical Code (IMC)
- International Energy Conservation Code (IECC)
- National Electrical Code (NEC)
- Applicable state and local building codes
- Manufacturer installation requirements

3.0 SCOPE OF WORK

3.1 HVAC Load Testing and System Sizing

- Perform detailed heating and cooling load calculations for each building
- Evaluate building envelope, insulation, occupancy and usage patterns.
- Determine required HVAC system capacities for optimal efficiency and performance
- Provide a sizing report

3.2 Removal of Existing Equipment

- Safely disconnect and remove all existing HVAC equipment, including
 - Air handling units
 - Condensers
 - Furnaces
 - Evaporative coolers
- Dispose of all equipment removed according to Forest Service Engineering.
- Recover and handle refrigerants per EPA requirements.

3.3 Installation of New HVAC Systems

- Furnish all install new high efficiency HVAC systems meeting current energy codes.
- Systems may include
 - Split systems
 - Packaged roof units
 - Heat pumps or variable refrigerant flow systems

- Install all associated components including thermostats, controls and filtration systems.
- Ensure proper system balancing and commissioning

3.4 Ductwork Repair and Replacement

- Inspect existing ductwork for leaks, damage, and inefficiencies
- Repair, seal, or replace ductwork as required
- Ensure duct systems meet airflow requirements and are properly insulated
- Test and balance airflow throughout the buildings.

3.5 Electrical Work

- Remove and replace all electrical components associated with HVAC systems needed.
- Install new electrical connections, disconnects, breakers, and control wiring.
- Ensure all electrical work complies with the National Electrical Code.
- Coordinate with existing building electric system for proper integration.

3.6 Decommissioning of Evaporative Cooling Systems

- Permanently disconnect and remove water supply lines associated with evaporative cooling systems.
- Cap, Seal, and secure all unused plumbing lines.
- Ensure no risk of leaks, contamination, or future system interference.

3.7 System Testing and Commissioning

- Perform full system startup and operational testing.
- Verify system performance meets design specifications.
- Conduct air balancing and temperature verification.
- Provide documentation of commissioning results.

4.0 GENERAL REQUIREMENTS

Equipment shall be properly sized for building square footage, intended use, and properly accommodate airflow needed according to the size of the building.

Contractor shall verify existing electrical service capacity and structural support prior to installation.

Work shall be coordinated to minimize disruption to facility operations.

5.0 LOCATION

Project work site is located at multiple facilities within the Smokey Bear Ranger District, Lincoln National Forest, 901 Mechem Drive, Ruidoso, NM, 88345.

See Attachment 3_MAP_FacilityRepair Project HVAC

See Attachment 4_MAP_Compound Buildings Numbered

Building names, numbers on the compound map, and approximate square footage:

BLDG #	BUILDING	APPROX. SQUARE FEET	SCOPE
1	Main Office	4,648 SF	HVAC replacement / upgrade
2	Small Office	916 SF	HVAC replacement / upgrade
3	Recreation Office	456 SF	HVAC replacement / upgrade
4	Fire Operations Building	4,125 SF	HVAC replacement / upgrade

5	Ranger House	1,704 SF	Remove evaporative cooling; install refrigerated HVAC
6	Tanker Bay	2,400 SF	Remove evaporative cooling; install refrigerated HVAC
7	Bunkhouse	840 SF	HVAC replacement / upgrade

Note: Square footages are approximate and provided for bidding purposes only.

6.0 GOVERNMENT-FURNISHED PROPERTY

The Government will provide access to the facilities.

No materials or equipment will be furnished by the Government.

7.0 QUALITY CONTROL AND INSPECTION

Contractor shall submit equipment submittals and electrical requirements prior to installation.

The Contracting Officer's Representative (COR) may inspect work at any stage.

Work not meeting contract requirements shall be corrected at no additional cost to the Government.

8.0 DELIVERABLES

- HVAC load calculation and sizing report
- Equipment submittals and specifications
- Installation
- Commissioning and testing
- Operation and maintenance manuals
- Warranty documentation
- Equipment data sheets and submittals
- Electrical load documentation
- Final inspection and commissioning documentation

9.0 PERFORMANCE PERIOD

Work shall be completed within 120 calendar days from Notice to Proceed (NTP).

10.0 SAFETY

The Contractor shall comply with all applicable OSHA requirements and current safety standards.

- Local building codes
- International Mechanical Code
- National Electrical Code
- ASHRAE standards
- EPA regulations for refrigerant hand

Contractor employees shall wear all proper Personal Protective Equipment (PPE) and follow safety measures during all phases of work.

The Contractor shall maintain a clean and safe job site.